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RESEARCH ARTICLE



# Ocean temperature reconstruction from satellite observations in the North Atlantic using an explainable deep learning framework

Fanli Liu<sup>a,b</sup>, Li Wei<sup>a,b,c</sup> and Lei Guan<sup>a,b,c</sup>

<sup>a</sup>Key Laboratory of Ocean Observation and Information of Hainan Province, Sanya Oceanographic Institution, Ocean University of China/SANYA Oceanographic Laboratory, Sanya, China; <sup>b</sup>College of Marine Technology, Faculty of Information Science and Engineering, Ocean University of China, Qingdao, China; <sup>c</sup>Laboratory for Regional Oceanography and Numerical Modeling, Qingdao Marine Science and Technology Center, Qingdao, China

## ABSTRACT

In this study, we reconstructed the North Atlantic Ocean temperature field by combining an interpretable deep learning framework with multi-source satellite data, including sea surface temperature (SST), sea surface height (SSH), sea level anomaly (SLA), sea surface wind field (SSW), and SST gradient (GRAD). The optimal model configuration achieved a root mean square error (RMSE) of 0.49 °C. This study extends the SHapley Additive exPlanations (SHAP) method to spatial, temporal, and vertical dimensions to systematically analyze the model's error sources and feature contributions. The results show that the reconstruction error is concentrated in regions with large physical gradients, such as the thermocline and Gulf Stream, exhibiting a significant seasonal bias pattern. SHAP analysis reveals a clear vertical transition in feature importance that SST dominates at all depths with a relative contribution of 34.60%. The contributions of SSW and GRAD were mainly concentrated at the sea surface, whereas the contributions of SSH and SLA began to increase in the lower mixed layer and peaked in the thermocline, becoming crucial in connecting the surface and ocean interior. This study highlights the depth and spatial dependence of feature contributions, providing a physical basis for the future optimization of remote sensing-based ocean temperature reconstruction models.

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## KEYWORDS

Ocean temperature reconstruction; explainable deep learning; sea surface observations; Shapley Additive Explanations

## 1. Introduction

The ocean is a fundamental component of the global climate system; it absorbs approximately 90% of excess heat and plays a crucial role in stabilising the climate system and maintaining the energy balance (Meyssignac et al. 2019; Von Schuckmann et al. 2020). Against the backdrop of global warming, accurate monitoring of the ocean's internal thermal structure is essential for understanding Earth's energy imbalances (Von Schuckmann et al. 2016) and heat redistribution (Cheng et al. 2017; Cheng et al. 2020; Yan et al. 2016). Three-dimensional (3D) ocean temperature fields are central to this monitoring. However, traditional in situ observations, such as the use of buoys, while highly accurate, are significantly limited with respect to spatial coverage. Conventional numerical ocean models are capable of generating 3D fields but are often constrained by high computational costs and complex initialisation requirements. These factors limit their utility for real-time applications. In contrast, satellite remote sensing can provide high-resolution, globally covered sea surface variables, such as sea surface temperature (SST), sea surface height (SSH), and sea surface wind (SSW). These observations significantly alleviate the issue of uneven and discontinuous spatial coverage in traditional datasets. Studies have shown that statistical correlations exist between sea surface observations and subsurface ocean temperature distributions, enabling the

**CONTACT** Li Wei ✉ [weili2068@ouc.edu.cn](mailto:weili2068@ouc.edu.cn) Key Laboratory of Ocean Observation and Information of Hainan Province, Sanya Oceanographic Institution, Ocean University of China/SANYA Oceanographic Laboratory, Sanya 572024, People's Republic of China; College of Marine Technology, Faculty of Information Science and Engineering, Ocean University of China, Qingdao 266100, People's Republic of China; the Laboratory for Regional Oceanography and Numerical Modeling, Qingdao, Marine Science and Technology Center, Qingdao 266061, People's Republic of China

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development of data-driven regression algorithms for reconstructing 3D ocean temperature fields (Fiedler 1988; Gilson et al. 1998; Klemas and Yan 2014; Willis, Roemmich, and Cornuelle 2004).

In recent years, data-driven methods represented by deep learning have made significant progress in various marine science tasks, such as internal wave detection (Zhang and Li 2024), underwater image captioning (Li et al. 2025; Li et al. 2026), and sea ice prediction (Ren et al. 2025), owing to their ability to effectively capture high-dimensional nonlinear relationships. In the field of ocean reconstruction, classic machine-learning models were widely used, such as fully connected neural networks (FCNNs) (Ali, Swain, and Weller 2004; Lu et al. 2019), self-organising map (SOM) (Wu et al. 2012), and multilayer perceptrons (MLPs) (Pauthenet et al. 2022; Sammartino et al. 2020). To capture more complex nonlinear relationships, subsequent research employed relatively powerful deep learning models including U-Net (Xie et al. 2022), convolutional LSTM (ConvLSTM) (Song et al. 2022; Su et al. 2022), various CNN architectures (Liu et al. 2024; Mao et al. 2023; Zhang et al. 2023a), ensemble models (Qi et al. 2022), and Transformer-based networks (Wang et al. 2024; Wu et al. 2024; Xie, Dong, and Xu 2025; Zhang et al. 2023b), thereby achieving improved accuracy and higher resolution.

However, the ultimate performance of a model depends not only on the algorithm but more crucially on the selected feature variables. Previous studies have shown that the SST, SSH, and SSW are closely related to the reconstruction of subsurface ocean temperatures. Ali, Swain, and Weller (2004) used SST, SSH, wind stress, net radiation, and net heat flux as inputs and found that SSH was the most important predictor. Lu et al. (2019) considered SST, SSH, and SSW and pointed out that SSH dominated the variation in subsurface temperature anomalies, whereas SSW moderately improved the reconstruction. Sensitivity tests by Sammartino et al. (2020) showed that incorporating absolute dynamic topography (ADT) substantially reduced reconstruction errors, with the root mean square error (RMSE) of temperature decreasing from 1.40 °C to 1.13 °C. Pauthenet et al. (2022) used sea level anomaly (SLA), mean dynamic topography (MDT), geostrophic velocity derived from MDT/SLA, and SST and found through SHapley Additive exPlanations (SHAP) analysis that SST was the dominant factor for shallow layers, whereas MDT became the most important variable below 200 m. Song et al. (2022) applied SST, sea surface salinity (SSS), SSH, and SSW and confirmed that incorporating SSH and SSW significantly reduced errors in the 30–200 m depth range. Smith et al. (2023) considered the SST, SSS, sea surface density, ADT, SLA, geostrophic currents, SSW, and spatiotemporal information. SHAP analysis indicated that SST and ADT were the main driving factors in the upper 200 m, with shallow layers mainly controlled by SST anomalies, whereas ADT became important between 100 and 600 m. Feng et al. (2025) used ADT, SST, SSW, latitude, longitude, and time to reconstruct the thermocline depth. The SHAP analysis showed that SST had the strongest influence, ADT also played a role, and SSW contributed mainly through wind-driven mixing, with weaker effects than those of SST and ADT.

Despite these advancements, several challenges persist. Most existing approaches focus on improving reconstruction performance from a modelling perspective, whereas systematic analyses of error sources and their relationships with input features are still limited. Furthermore, selecting a representative region with complex dynamics is crucial for validating the efficacy of deep learning models. In this study, we proposed an explainable deep learning framework specifically designed to reconstruct a 3D ocean temperature field. We focused on the North Atlantic Ocean as our study area. This region constitutes the Gulf Stream, Atlantic Meridional Overturning Circulation (AMOC), and energetic mesoscale eddy activities, offering a diverse range of oceanographic conditions from coastal shelves to the deep open ocean. The successful reconstruction of the 3D temperature field in such a dynamically complex region serves as a rigorous benchmark for verifying the physical consistency and generalisation capabilities of the model. The main contribution of this study is the quantitative elucidation of how sea surface satellite signatures physically drive the reconstruction of subsurface temperatures at different depths. Particularly, by integrating the SHAP analysis with spatial, temporal, and vertical dimensions, we systematically dissected the distinct roles of surface features in the reconstruction process. This multidimensional analysis reveals two key physical insights: 1) reconstruction errors are not randomly distributed but are structurally concentrated in high-gradient regions, such as the thermocline and Gulf Stream; and 2) the contributions of surface features exhibit a clear vertical transition, while the influence of SST extends to deeper layers. SSW and SST gradient (GRAD) are prominent on the surface, and SSH and SLA play a crucial role within the thermocline, effectively bridging surface signals to the ocean interior. These findings provide a solid

**Table 1.** Seventy-five depth layers of the ocean temperature product (European Union-Copernicus Marine Service 2021).

| Layer 1–10 | Layer 11–20 | Layer 21–30 | Layer 31–40 | Layer 41–50 | Layer 51–60 | Layer 61–70 | Layer 71–75 |
|------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| 0.51 m     | 16.53 m     | 69.02 m     | 199.79 m    | 565.29 m    | 1516.36 m   | 3138.57 m   | 5089.48 m   |
| 1.56 m     | 19.43 m     | 77.61 m     | 221.14 m    | 628.03 m    | 1652.57 m   | 3324.64 m   | 5291.68 m   |
| 2.67 m     | 22.76 m     | 86.93 m     | 244.89 m    | 697.26 m    | 1795.67 m   | 3513.45 m   | 5494.58 m   |
| 3.86 m     | 26.56 m     | 97.04 m     | 271.36 m    | 773.37 m    | 1945.30 m   | 3704.66 m   | 5698.06 m   |
| 5.14 m     | 30.87 m     | 108.03 m    | 300.89 m    | 856.68 m    | 2101.03 m   | 3897.98 m   | 5902.06 m   |
| 6.54 m     | 35.74 m     | 120.00 m    | 333.86 m    | 947.45 m    | 2262.42 m   | 4093.16 m   |             |
| 8.09 m     | 41.18 m     | 133.08 m    | 370.69 m    | 1045.85 m   | 2429.03 m   | 4289.95 m   |             |
| 9.82 m     | 47.21 m     | 147.41 m    | 411.79 m    | 1151.99 m   | 2600.38 m   | 4488.16 m   |             |
| 11.77 m    | 53.85 m     | 163.16 m    | 457.63 m    | 1265.86 m   | 2776.04 m   | 4687.58 m   |             |
| 13.99 m    | 61.11 m     | 180.55 m    | 508.64 m    | 1387.38 m   | 2955.57 m   | 4888.07 m   |             |

physical foundation for future research, particularly for optimising the feature selection for depth-specific reconstruction models.

The remainder of this paper is organised as follows: [Section 2](#) details the data sources and study area; [Section 3](#) describes the interpretable deep learning framework used; [Section 4](#) presents the reconstruction results; [Section 5](#) provides a comprehensive accuracy assessment and error analysis of the results; and [Section 6](#) summarises the main conclusions.

## 2. Data

### 2.1. Sea surface variables

The sea surface data used as inputs in this study cover key ocean parameters representing the major physical processes regulating ocean heat distribution, including SST, SSH, SLA, SSW, and GRAD. SST directly reflects the thermal state of the ocean surface and is a key driver of air–sea heat flux (Cayan 1992). SSH and SLA jointly describe the ocean’s dynamic processes. The SSH reflects large-scale quasi-geostrophic circulation such as the Gulf Stream (Wunsch and Stammer 1998). In contrast, the SLA captures transient mesoscale features that deviate from the mean state (Chelton, Schlax, and Samelson 2011). These features are the key mechanisms for vertical heat transport and energy mixing in the ocean (Toggweiler and Russell 2008). SSW influences mixed layer thickness and momentum transfer through wind-driven mixing and by driving Ekman transport (Ekman 1905; Large, McWilliams, and Doney 1994). GRAD, when correlated with SST, captures the local horizontal variability that SST alone cannot fully represent. Specifically, it identifies ocean fronts (Belkin, Cornillon, and Sherman 2009). These frontal features are closely related to local circulation and dynamic instabilities.

The SST data were obtained from the Copernicus Marine Data Store, named the Global Ocean OSTIA Sea Surface Temperature and Sea Ice Reprocessed. Generated by the Operational SST and Ice Analysis (OSTIA) system using satellite and in situ foundation SST data (European Union-Copernicus Marine Service 2015), this product provides a daily average at a  $0.05^\circ$  resolution from 1981 to the present. The SSH and SLA data were obtained from the Copernicus Marine Data Store, named as Global Ocean Gridded L4 Sea Surface Heights and Derived Variables Reprocessed 1993–Ongoing product. Derived from satellite altimeter missions (European Union-Copernicus Marine Service 2021), this product provides a daily average at  $0.25^\circ$  resolution, covering the period from 1993 to the present. The SSW data were obtained from the Remote Sensing Systems called Cross-Calibrated Multi-Platform (CCMP) Wind Vector Analysis Product. By combining 10-m wind fields from various satellite microwave sensors with reanalysis background fields (Mears et al. 2022), this product has a temporal resolution of 6 h and spatial resolution of  $0.25^\circ$  from 1993 to the present.

### 2.2. 3D Ocean temperature

The ocean temperature data used as the ground truth in this study were obtained from the Global Ocean Ensemble Physics Reanalysis of the Copernicus Marine Data Store (European Union-Copernicus Marine Service 2019). This product was generated by a numerical ocean model constrained by data assimilation from satellites and in situ observations, providing a physically consistent and gap-free 3D ocean state estimate. This product provides daily averages at a  $0.25^\circ$  resolution and covers the period from 1993 to the

present. It encompasses 75 vertical levels, ranging in depth from approximately 0.5 m to nearly 6000 m, as shown in Table 1. This product exhibits high accuracy, with a global mean bias of less than 0.1 °C and a root mean square difference of approximately 0.49 °C averaged over the full 0–5000 m (Bourdallé-Badie and Carabin, 2024).

### 2.3. Study area and data preprocessing

The study area was the North Atlantic, spanning 100°W to 10°E and 5°S to 65°N. Characterised by the Gulf Stream, AMOC, and frequent mesoscale eddy activity, the North Atlantic is a key region for global energy transport and temperature regulation. The mixed layer depth (MLD) in the North Atlantic exhibited strong seasonal and regional variations. In winter, low temperatures, strong winds, and enhanced surface heat loss lead to significant deepening of the mixed layer. In summer, increased solar radiation, weaker winds, and longer sunshine hours increase SSTs and strengthen stratification, resulting in a relatively shallow mixed layer. Spatially, the mixed layer remains deep throughout the year in high-latitude regions, such as the Labrador Sea, reaching 700 m in winter. Conversely, tropical regions generally have shallow mixed layers with small seasonal variations. This complex vertical structure renders 3D temperature reconstruction particularly challenging. The MLD data used in this study were obtained from the Global Ocean Ensemble Physics Reanalysis product.

We collected data from 2012 to 2020 to construct the dataset. Data from 2019 to 2020 were used as the test set to evaluate model performance. The remaining data from 2012 to 2018 were divided into training and validation sets at a ratio of 9:1. To ensure physical consistency across multi-source data, several preprocessing steps were implemented to address spatiotemporal mismatches. First, regarding spatial alignment, we identified the primary sources of spatial mismatch in both the resolution and grid-centre definitions. To resolve this, we established a unified pixel-centred target grid as  $G_{target}$  based on the 0.25° resolution of ocean temperature data:

$$G_{target} = \{(\phi_i, \lambda_j) \mid \phi_i = 0.125^\circ \cdot (2i + 1), \lambda_j = 0.125^\circ \cdot (2j + 1)\} \quad (1)$$

where  $i$  and  $j$  are integers indexing the latitude and longitude, respectively. All input variables were mapped onto  $G_{target}$  using bilinear interpolation. Particularly, the SST at 0.05° resolution was down-sampled, and variables defined on the grid corners were realigned to the target pixel centres, such as the SSW and ocean temperature data. The value of the target grid  $V_t$  is computed as follows:

$$V_t(\phi, \lambda) = \sum_{k=1}^4 w_k \cdot V_s(x_k, y_k) \quad (2)$$

where  $(x_k, y_k)$  are the nearest neighbours in the source grid and  $w_k$  represents the distance-weighted coefficients. Second, to resolve the temporal mismatch, the 6-h SSW data were aggregated on a daily scale using the following arithmetic mean:

$$\bar{S}_{daily} = \frac{1}{4} \sum_{t=0}^3 S_{6h}(t) \quad (3)$$

This ensures that dynamic wind forcing is temporally synchronised with the daily average ocean temperature data. Following spatiotemporal alignment, the magnitude of the spatial gradient of the SST (GRAD) was calculated to capture the frontal features, which is expressed as follows:

$$GRAD = \sqrt{\left(\frac{\partial SST}{\partial x}\right)^2 + \left(\frac{\partial SST}{\partial y}\right)^2} \quad (4)$$

where  $x$  and  $y$  represent zonal and meridional directions, respectively. Finally, all input and target variables were standardised using Z-score normalisation to remove the influence of different physical scales and accelerate model convergence.

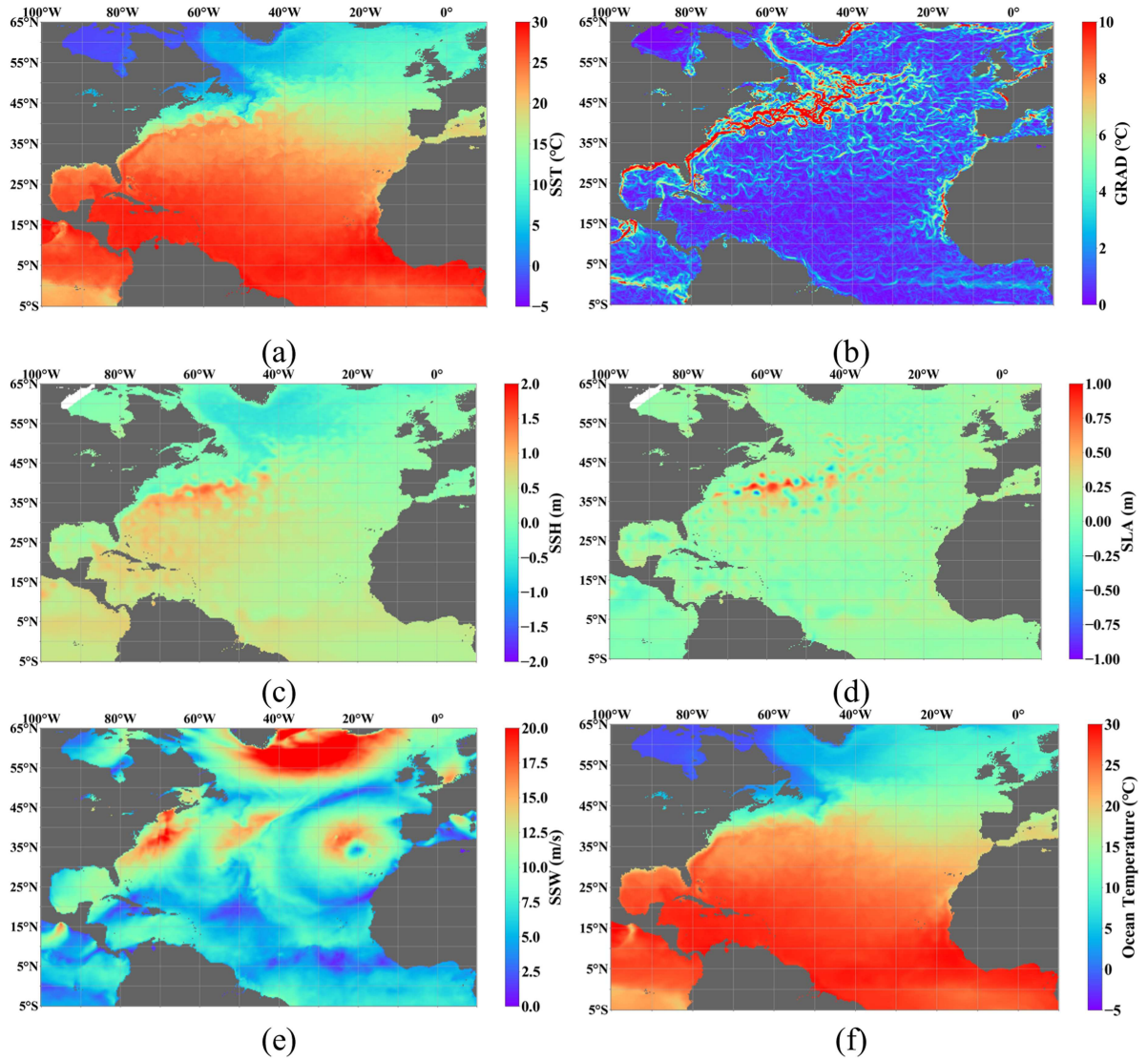
$$X_{norm} = \frac{X - \mu}{\sigma} \quad (5)$$

where  $X$  represents the original data, and  $\mu$  and  $\sigma$  are the mean and standard deviation (STD) of the dataset, respectively. Figure 1 shows the sea surface variables and ocean temperature at a depth of 0.5 m on December 1, 2020.

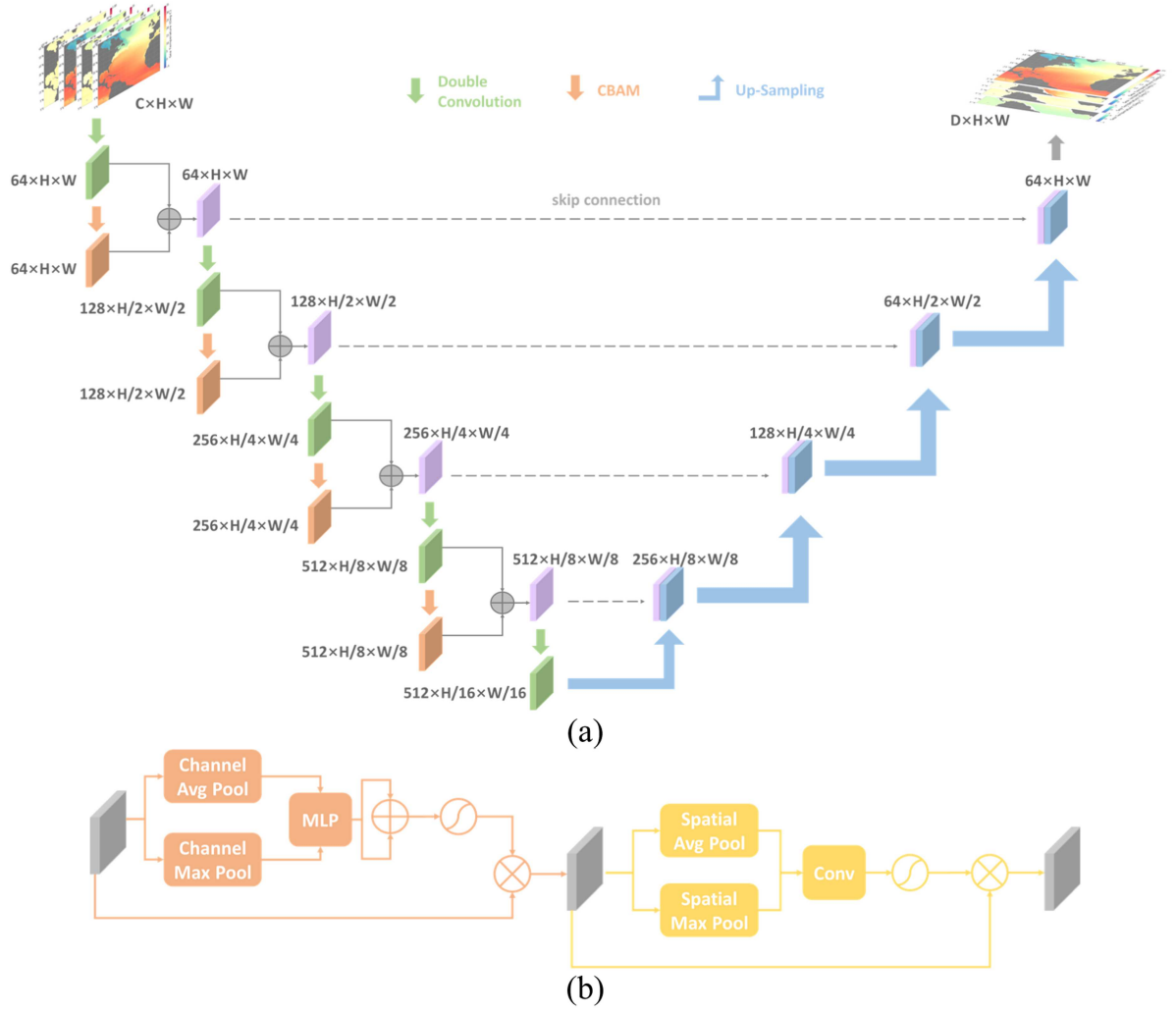
### 3. Method

#### 3.1. Deep learning model

The deep learning model utilised in this study combined the U-Net (Ronneberger, Fischer, and Brox 2015) architecture with a Convolutional Block Attention Module (CBAM) (Woo et al. 2018) to enhance the performance of 3D ocean temperature reconstruction. The architecture of the model is shown in Figure 2. U-Net is a deep convolutional neural network renowned for its symmetric encoder–decoder structure. The encoder progressively extracts high-level semantic features through a series of downsampling operations, including convolution and pooling. Conversely, the decoder gradually restores spatial resolution through



**Figure 1.** Sample from the North Atlantic dataset on December 1, 2020. (a) SST, (b) GRAD, (c) SSH, (d) SLA, (e) SSW, and (f) ocean temperature at a depth of 0.5 m. Grey areas represent land.



**Figure 2.** Model architecture. (a) the overall architecture. (b) the CBAM.

upsampling operations, including deconvolution. A key characteristic of U-Net is the use of skip connections that pass low-level spatial information from the encoder layers to the corresponding decoder layers, thereby preventing the loss of fine-grained details in the deep network.

Although U-Net effectively captures multiscale features, it may treat all features with equal importance. To address this limitation, a CBAM module was introduced to improve the feature selection capabilities of the model. CBAM enhances U-Net by sequentially inferring attention weights along the channel and spatial dimensions. This allows the model to learn what should be emphasised at the channel level and where to focus at the spatial level.

The Channel Attention module identifies the most informative feature channels. It first aggregates the spatial information from the input feature map  $F$  using global average pooling (*AvgPool*) and global max pooling (*MaxPool*). Both descriptors were processed using a shared MLP. The resulting feature vectors were merged using an element-wise summation and subsequently passed through a sigmoid activation function ( $\sigma$ ) to generate the final channel attention map  $M_c$ . The computation is as follows:

$$M_c(F) = \sigma(\text{MLP}(\text{AvgPool}(F)) + \text{MLP}(\text{MaxPool}(F))) \quad (6)$$

This attention map  $M_c$  was then multiplied by the input feature map  $F$  to recalibrate the channel-wise features.

The Spatial Attention module focuses on identifying the most salient spatial regions. It uses the channel-refined feature map  $F'$  and applies average and maximum pooling operations along the channel axis. These two outputs were concatenated to create an effective two-channel feature descriptor. This descriptor was then processed using a standard  $7 \times 7$  convolutional layer ( $f^{7 \times 7}$ ), followed by a sigmoid activation function to produce the spatial attention map  $M_s$ :

$$M_s(F') = \sigma(f^{7 \times 7}([\text{AvgPool}(F'); \text{MaxPool}(F')])) \quad (7)$$

This map  $M_s$  was multiplied with  $F'$  to spatially refine the features. The CBAM is integrated into the network using a residual connection strategy. The attention-refined features are element-wise added to the original input features. This residual design facilitates gradient flow and prevents the loss of original feature information during the attention refinement process. By incorporating the CBAM into each convolutional block of U-Net, the model enhances its ability to extract and represent different features, enabling it to focus on the most relevant spatial and channel-specific information during the reconstruction process.

Crucially, to ensure the vertical consistency of the reconstructed temperature fields, we employed a unified modelling strategy rather than training separate models for each depth. The final layer of the decoder utilised a  $1 \times 1$  convolution to project high-dimensional features onto 75 vertical levels of the ocean. In contrast with layer-wise approaches that model each depth independently, this end-to-end architecture enables the network to explicitly learn vertical connections via channel-wise dependencies. By sharing the same encoder-decoder pathway, the model captures the inherent thermodynamic coupling of the water column, thereby ensuring physically consistent vertical profiles.

To train the model, we employed a Masked Loss function designed to handle irregular ocean boundaries. Given that the input data contain invalid land mask values, simply calculating the error over the entire grid introduces a significant bias. Therefore, we applied a binary mask  $M$  to exclude the land regions from the calculations. Furthermore, to encourage the model to accurately reconstruct the entire vertical profile simultaneously, the loss was calculated by summing the RMSE across all 75 vertical levels rather than averaging them. This summation strategy ensures that errors in specific layers contribute significantly to the total gradient, thereby preventing dilution. The loss function  $L$  is expressed as follows:

$$L = \sum_{d=1}^D \sqrt{\frac{1}{N_d} \sum_{i,j} M_{i,j} (Y_{d,i,j} - Y'_{d,i,j})^2} \quad (8)$$

where  $D = 75$  is the number of depth levels;  $N_d$  is the number of valid ocean pixels at depth  $d$ ;  $Y_{d,i,j}$  and  $Y'_{d,i,j}$  denote the ground truth and predicted temperature at depth  $d$ , latitude  $i$ , and longitude  $j$ , respectively.  $M_{i,j}$  is a binary land mask, where 1 represents the ocean and 0 represents land. By minimising this summed volumetric loss, the model was optimised to reduce the errors across the entire water column.

### 3.2. SHAP

SHAP (Lundberg and Lee 2017) is a model interpretability method based on the cooperative game theory. This method is based on Shapley values, which are used to fairly distribute collective gains among multiple cooperative participants. SHAP applies this concept to machine learning to compute the contribution of each feature to the individual predictions. SHAP values are calculated by evaluating the model output for all possible feature combinations. It calculates the marginal contribution of a feature by measuring the change in the model predictions when that feature is added to a subset of other features. This calculation is weighted and summed over all possible subsets, as shown in the following formula:

$$\phi_j(f) = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{j\}) - f(S)] \quad (9)$$

where  $\phi_j(f)$  represents the Shapley value of feature  $j$ ,  $N$  is the set of all features,  $N \setminus \{j\}$  is the set of features excluding feature  $j$ , and  $S$  is the set of all subsets that do not contain feature  $j$ .  $f(S)$  is the model's output

for the feature subset  $S$ , and  $f(S \cup \{j\})$  is the model's output when feature  $j$  is added to subset  $j$ . The coefficient  $\frac{|S|! (|N| - |S| - 1)!}{|N|!}$  is the weight, indicating the influence of feature  $j$  when added to a subset.

The SHAP value can be positive or negative. A positive value indicates that the feature contributes positively by increasing the predicted ocean temperature, whereas a negative value implies a negative contribution, decreasing the prediction. The absolute magnitude of the SHAP value reflects the strength of the contribution of the feature, with larger absolute values signifying a greater influence on prediction. Using this method, the impacts of surface features on ocean temperature reconstruction can be quantitatively assessed, revealing the contributions of different input features at various depths.

## 4. Results

### 4.1. Comparative performance and stability analysis

To comprehensively evaluate the reconstruction capability of the UNet-CBAM, we conducted a comparison with several established deep learning architectures, including the standard U-Net (Ronneberger, Fischer, and Brox 2015), deep residual network ResNet-50 (He et al. 2016), and DeepLabV3+ (Chen et al. 2018). Furthermore, considering the rise of Vision Transformer (ViT) (Dosovitskiy et al. 2021) in computer vision, we also compared with a pure Transformer-based architecture and a hybrid architecture (UTrans) that combines the advantages of CNN and Transformer. All the models were trained and tested under identical experimental conditions. As presented in Table 2, the UNet-CBAM achieved the best performance, with the lowest RMSE of 0.49 °C, significantly outperforming the other comparative baselines. Among the CNN baseline models, ResNet-50 performed moderately (RMSE 0.5884 °C). The deep downsampling operations in ResNet resulted in significant loss of spatial resolution, and the lack of a symmetrical decoder path hindered the effective recovery of these lost spatial details. Although DeepLabV3+ introduced Atrous Spatial Pyramid Pooling (ASPP) to capture multi-scale context, its error was larger (RMSE 0.7504 °C). This may be because atrous convolutions can introduce unnecessary grid artefacts when processing continuous physical field regression tasks requiring high fidelity and smooth gradients, leading to decreased reconstruction accuracy. Notably, the pure Transformer model performed the worst (RMSE 0.9097 °C), while the hybrid architecture UTrans showed significant performance improvement (RMSE 0.5016 °C), but still slightly lagged behind the U-Net-based method. This performance gap is mainly attributed to the inductive bias of the model architecture. Pure Transformer models divide the input into flattened image patches, which disrupts the crucial local spatial continuity in the ocean temperature field, making it difficult for the model to reconstruct high-frequency details such as the bottom of the mixed layer or frontal edges. In contrast, the inherent translation invariance and local receptive field of CNNs make them more suitable for such dense regression tasks. Although UTrans mitigates this problem by incorporating a CNN encoder, UNet-CBAM demonstrates higher data efficiency with the dataset size used in this study. U-Net's skip connections directly preserve high-resolution features from shallow layers, while the CBAM module further suppresses irrelevant features, thus achieving an optimal balance between capturing global context and preserving fine local structures.

In addition to inter-model comparison, verifying the necessity of the attention mechanism and statistical reliability of the results is crucial. To this end, we conducted a rigorous ablation study coupled with a stability analysis by training both Standard U-Net and UNet-CBAM five times using different random seeds. The statistical results summarised in Table 3 convincingly demonstrate that the inclusion of the CBAM module provides a consistent performance gain. The average RMSE decreased from 0.5073 °C

**Table 2.** Comparison of RMSE, bias, and  $R^2$  for different models under identical experimental conditions.

| Model       | RMSE (°C) | Bias (°C) | $R^2$  |
|-------------|-----------|-----------|--------|
| UNet-CBAM   | 0.4921    | -0.0063   | 0.9969 |
| U-Net       | 0.5137    | 0.0174    | 0.9967 |
| ResNet      | 0.5884    | 0.0176    | 0.9956 |
| DeeplabV3+  | 0.7504    | -0.0723   | 0.9929 |
| Transformer | 0.9097    | 0.0029    | 0.9895 |
| UTrans      | 0.5016    | -0.0318   | 0.9968 |

**Table 3.** Ablation study and stability analysis. Comparison of reconstruction performance between the Standard U-Net and proposed UNet-CBAM. The statistics (Mean  $\pm$  STD) are calculated from five independent training runs with different random seeds.

| Model        | RMSE ( $^{\circ}\text{C}$ ) | Bias ( $^{\circ}\text{C}$ ) | $R^2$               |
|--------------|-----------------------------|-----------------------------|---------------------|
| With CBAM    | $0.4969 \pm 0.0036$         | $0.0056 \pm 0.0224$         | $0.9969 \pm 0.0001$ |
| Without CBAM | $0.5073 \pm 0.0075$         | $0.0218 \pm 0.0112$         | $0.9967 \pm 0.0001$ |

for the Standard U-Net to  $0.4969^{\circ}\text{C}$  for the UNet-CBAM, representing a robust improvement in reconstruction accuracy across all trials. Furthermore, the proposed method exhibits significantly enhanced training stability. The STD of the RMSE for UNet-CBAM is  $0.0036^{\circ}\text{C}$ , which is less than half of that observed for the Standard U-Net ( $0.0075^{\circ}\text{C}$ ). This reduced variance indicates that the attention mechanism effectively suppresses irrelevant features and noise, assisting the model in converging to a robust solution regardless of the weight initialisation.

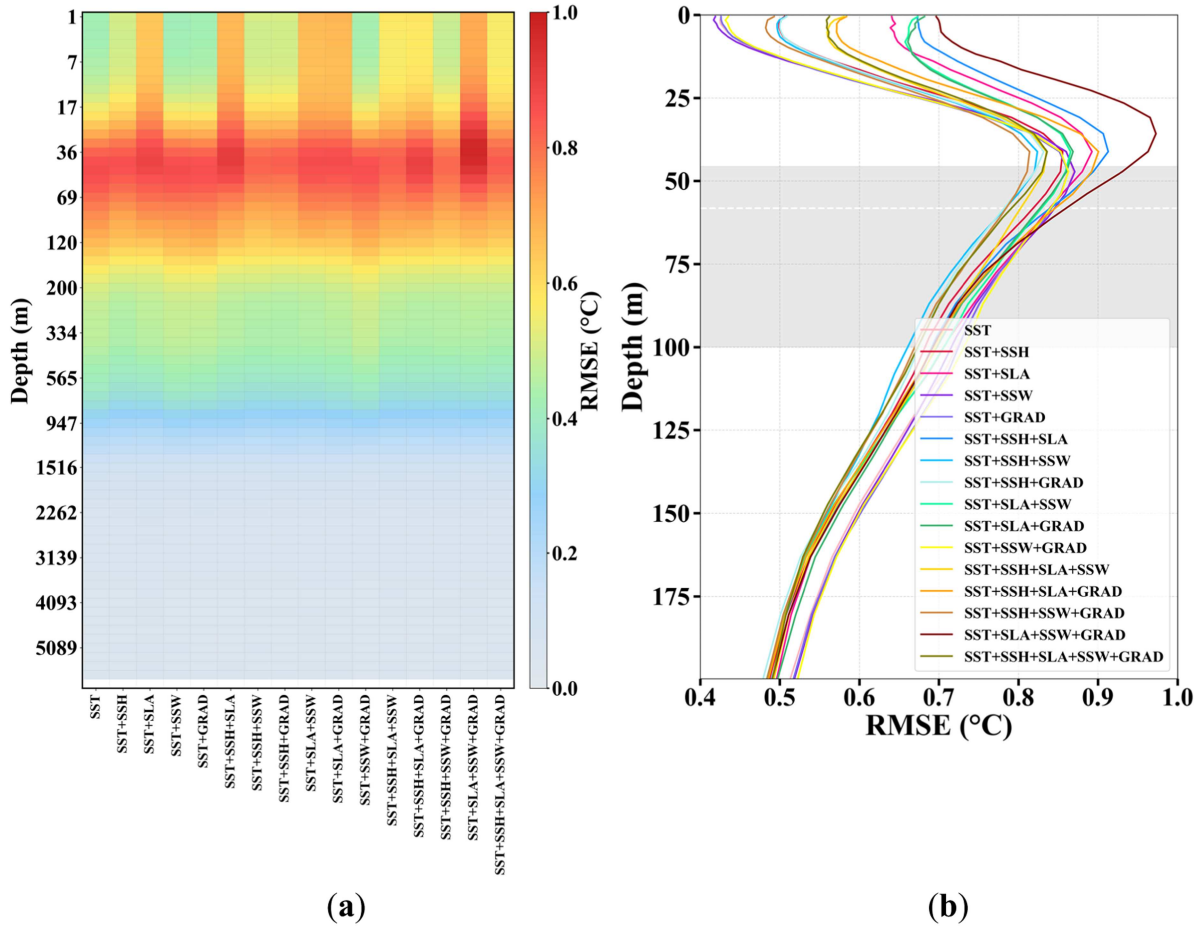
#### 4.2. Impact of different input feature combinations

To determine the optimal input features, we evaluated the performance of the models trained with different feature combinations. Each model was trained independently for 300 epochs, and the model checkpoint with the lowest validation loss was selected as the final optimal model for testing, to ensure that the model maintained its optimal generalisation capability. Table 4 presents a summary of the quantitative performance metrics, specifically the RMSE for the mixed layer, thermocline, and entire depth, as well as the maximum RMSE and related depth. Figure 3 shows the vertical profile of the RMSE at 75 depth layers for different combinations. Figure 3(a) illustrates the overall error trend from 0.5 m to approximately 6000 m. All feature combinations exhibit a similar vertical structure where the RMSE increases in the shallow water, peaks near the thermocline boundary, decreases after the peak, and then decreases sharply below 1500 m, gradually approaching  $0^{\circ}\text{C}$ . This profile is consistent with the ocean's physical stratification, reflecting the dynamic upper ocean, including the mixed layer and thermocline, and a relatively calm deep ocean. Figure 3(b) focuses on the upper 200 m, highlighting that the peak RMSE for all models occurs near the top of the mean thermocline, as indicated by the grey shaded region. As shown in Table 4, the error within the thermocline ranging from  $0.89^{\circ}\text{C}$  to  $1.00^{\circ}\text{C}$  is substantially higher than that in the mixed layer ranging from  $0.43^{\circ}\text{C}$  to  $0.69^{\circ}\text{C}$ , confirming that the steep vertical temperature gradient remains the primary challenge for this reconstruction task. Furthermore, the overall average Coefficient of Determination ( $R^2$ ) is 0.99, confirming that the model successfully captured the vast majority of the variance in the ocean temperature field.

Table 4 and Figure 3 also present the specific impacts of different features. The inclusion of the SLA generally had a negative impact on the model performance. Combinations including the SLA consistently

**Table 4.** RMSE statistics for multiple feature combination models in the mixed layer, thermocline, and all depths, along with the maximum RMSE values and depths at which these maximum RMSE values occur.

| Sea surface features     | RMSE <sub>Mix</sub> ( $^{\circ}\text{C}$ ) | RMSE <sub>Thermocline</sub> ( $^{\circ}\text{C}$ ) | RMSE <sub>All depth</sub> ( $^{\circ}\text{C}$ ) | RMSE <sub>max</sub> ( $^{\circ}\text{C}$ ) | Depth of RMSE <sub>max</sub> (m) |
|--------------------------|--------------------------------------------|----------------------------------------------------|--------------------------------------------------|--------------------------------------------|----------------------------------|
| SST                      | 0.43                                       | 0.96                                               | 0.50                                             | 0.87                                       | 47.21                            |
| SST, SSH                 | 0.50                                       | 0.92                                               | 0.51                                             | 0.86                                       | 41.18                            |
| SST, SLA                 | 0.62                                       | 0.96                                               | 0.55                                             | 0.89                                       | 41.18                            |
| SST, SSW                 | 0.43                                       | 0.96                                               | 0.51                                             | 0.87                                       | 47.21                            |
| SST, GRAD                | 0.43                                       | 0.96                                               | 0.50                                             | 0.86                                       | 47.21                            |
| SST, SSH, SLA            | 0.65                                       | 0.96                                               | 0.56                                             | 0.91                                       | 41.18                            |
| SST, SSH, SSW            | 0.48                                       | 0.90                                               | 0.50                                             | 0.82                                       | 41.18                            |
| SST, SSH, GRAD           | 0.49                                       | 0.91                                               | 0.50                                             | 0.83                                       | 41.18                            |
| SST, SLA, SSW            | 0.62                                       | 0.95                                               | 0.55                                             | 0.87                                       | 41.18                            |
| SST, SLA, GRAD           | 0.63                                       | 0.94                                               | 0.55                                             | 0.87                                       | 41.18                            |
| SST, SSW, GRAD           | 0.44                                       | 0.96                                               | 0.51                                             | 0.86                                       | 47.21                            |
| SST, SSH, SLA, SSW       | 0.54                                       | 0.91                                               | 0.52                                             | 0.84                                       | 41.18                            |
| SST, SSH, SLA, GRAD      | 0.56                                       | 0.95                                               | 0.53                                             | 0.90                                       | 41.18                            |
| SST, SSH, SSW, GRAD      | 0.48                                       | 0.89                                               | 0.49                                             | 0.81                                       | 41.18                            |
| SST, SLA, SSW, GRAD      | 0.69                                       | 1.00                                               | 0.58                                             | 0.97                                       | 35.74                            |
| SST, SSH, SLA, SSW, GRAD | 0.54                                       | 0.91                                               | 0.51                                             | 0.84                                       | 41.18                            |



**Figure 3.** Variation of RMSE of the temperature reconstruction errors with depth for multiple feature combination models. (a) All depths. (b) Depth range of 0–200 m. Different colours in (a) represent RMSE values. Different colours in (b) represent different feature combinations. The grey shaded area in (b) represents the average upper and lower boundaries of the thermocline.

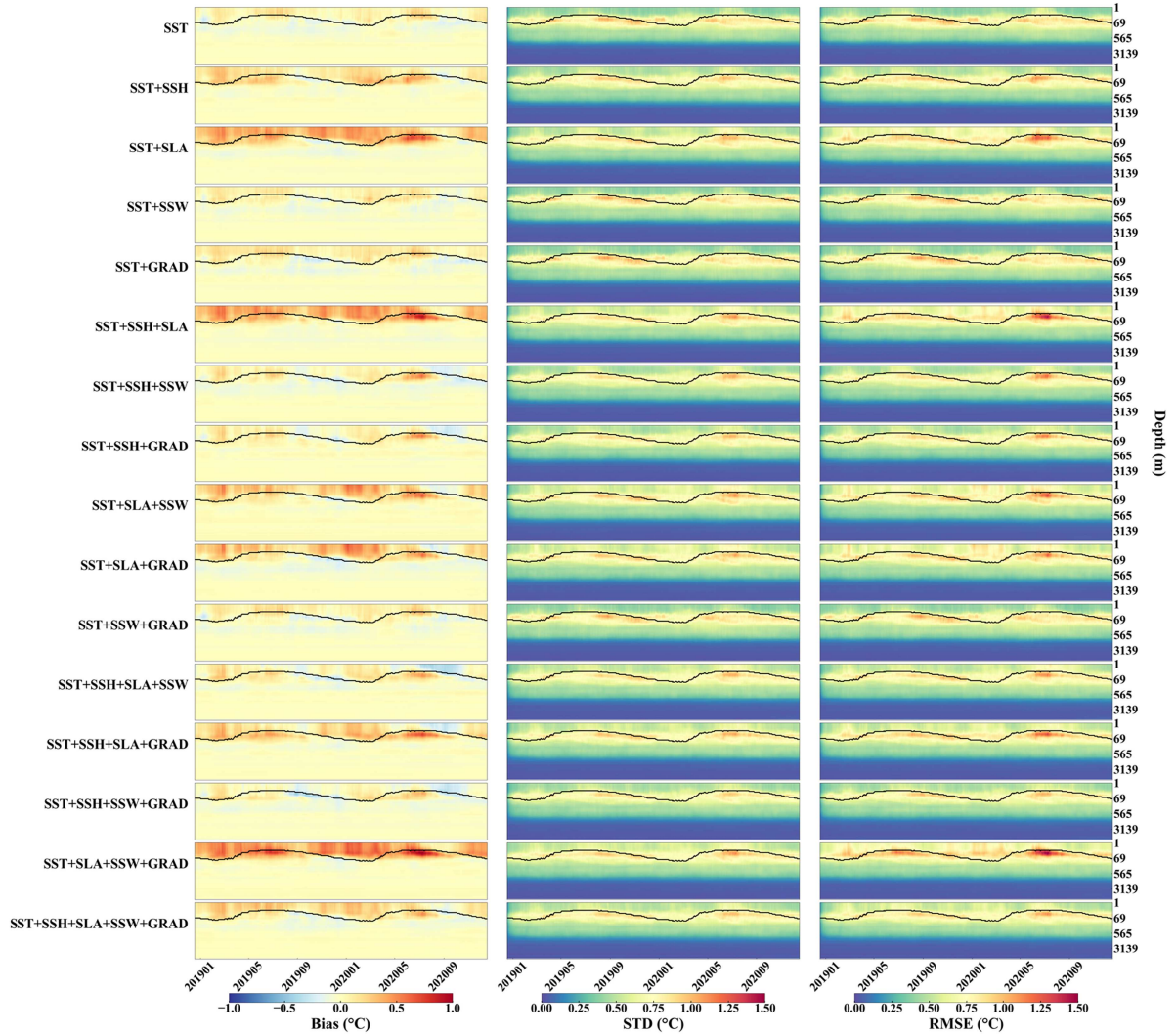
reduced accuracy, particularly in the mixed layer. For instance, adding SLA to the model using only SST increased the RMSE in the mixed layer from  $0.43^{\circ}\text{C}$  to  $0.62^{\circ}\text{C}$  without providing any improvement in the thermocline. This suggests that the high-frequency mesoscale variability contained in the SLA introduces significant nonlinearity and noise that disrupts the reconstruction of the relatively stable mixed layer. In contrast, SSH, which represents the MDT combined with anomalies, acts as a critical constraint on the subsurface structure. The inclusion of SSH effectively reduced the error in the challenging thermocline layer, lowering the RMSE to  $0.92^{\circ}\text{C}$  compared with  $0.96^{\circ}\text{C}$  in the model using only SST. SSW and GRAD played supplementary roles. Although their individual inclusion into the SST model showed minimal improvement in overall RMSE, they contributed to stabilising the model in the multi-feature configuration, specifically by reducing the maximum RMSE to a minimum value of  $0.81^{\circ}\text{C}$ . Furthermore, the inclusion of dynamic features such as SSH and SLA shifted the depth of maximum RMSE from 47.21 m to 41.18 m, indicating that the dynamically constrained models are more sensitive to near-surface variations and tend to resolve structures with steeper gradients at a slightly shallower depth.

The evaluation of the optimal model requires balancing the conflicting metrics listed in Table 2. While the model using only SST achieved the best performance in the mixed layer with an RMSE of  $0.43^{\circ}\text{C}$ , it failed to effectively constrain the thermocline, yielding a high RMSE of  $0.96^{\circ}\text{C}$ . Given that the thermocline has the highest error magnitude and physical complexity, we identified the reconstruction accuracy within the thermocline and the performance across the entire depth as the decisive standard. Thus, a combination of SST, SSH, SSW, and GRAD was selected as the optimal model. Although its mixed layer RMSE of  $0.48^{\circ}\text{C}$  is slightly higher than that of the SST model, it achieves the lowest overall RMSE of  $0.49^{\circ}\text{C}$ , lowest

thermocline RMSE of  $0.89^{\circ}\text{C}$ , and lowest maximum RMSE of  $0.81^{\circ}\text{C}$ . This confirms that SSH provides essential dynamic constraints for the subsurface, whereas SSW and GRAD offer beneficial boundary information that suppresses extreme errors, collectively producing the most robust reconstruction performance.

#### 4.3. Seasonal and spatial validation

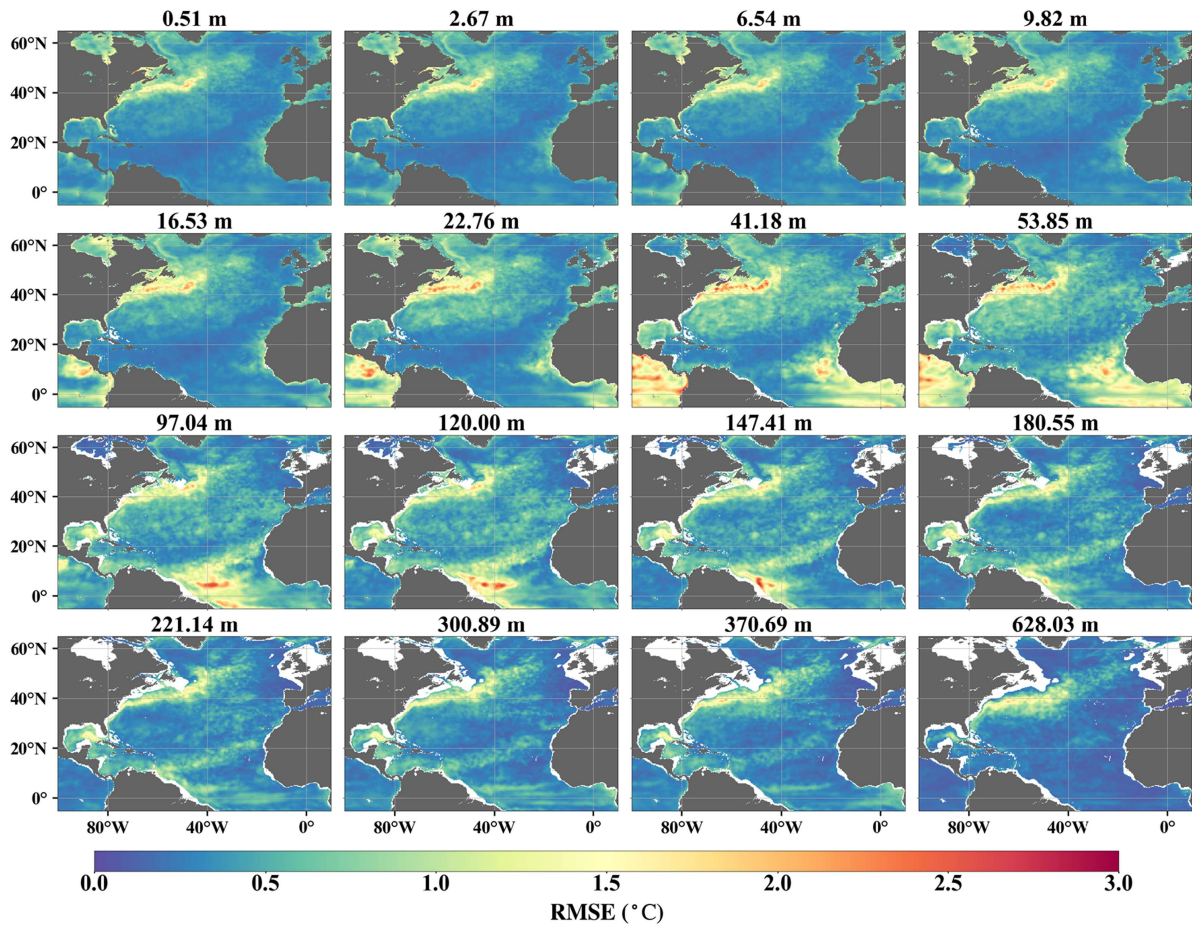
Figure 4 shows the variations in RMSE, mean bias, and STD with depth and time. The x-axis represents the data, y-axis represents the depth, colour indicates the magnitude of the error, and solid black line represents the depth of the mixed layer. The main variation in error was concentrated in the upper ocean from 0 to 500 m. Below 500 m, the error stabilised rapidly with increasing depth, RMSE decreased from approximately  $0.5^{\circ}\text{C}$  to near  $0^{\circ}\text{C}$ , and bias remained near  $0^{\circ}\text{C}$ , showing little seasonal or temporal variation. In contrast, the error in the 0–200 m depth range exhibited strong depth dependence and seasonal variation. For RMSE, significant errors were concentrated at the boundary between the mixed layer and thermocline, approaching  $1^{\circ}\text{C}$ . Larger errors exceeding  $1^{\circ}\text{C}$  appeared in the upper thermocline during late summer and autumn. This indicates that the thermocline, as the region with the most significant temperature gradient, amplified the reconstruction error of the model because of its steep



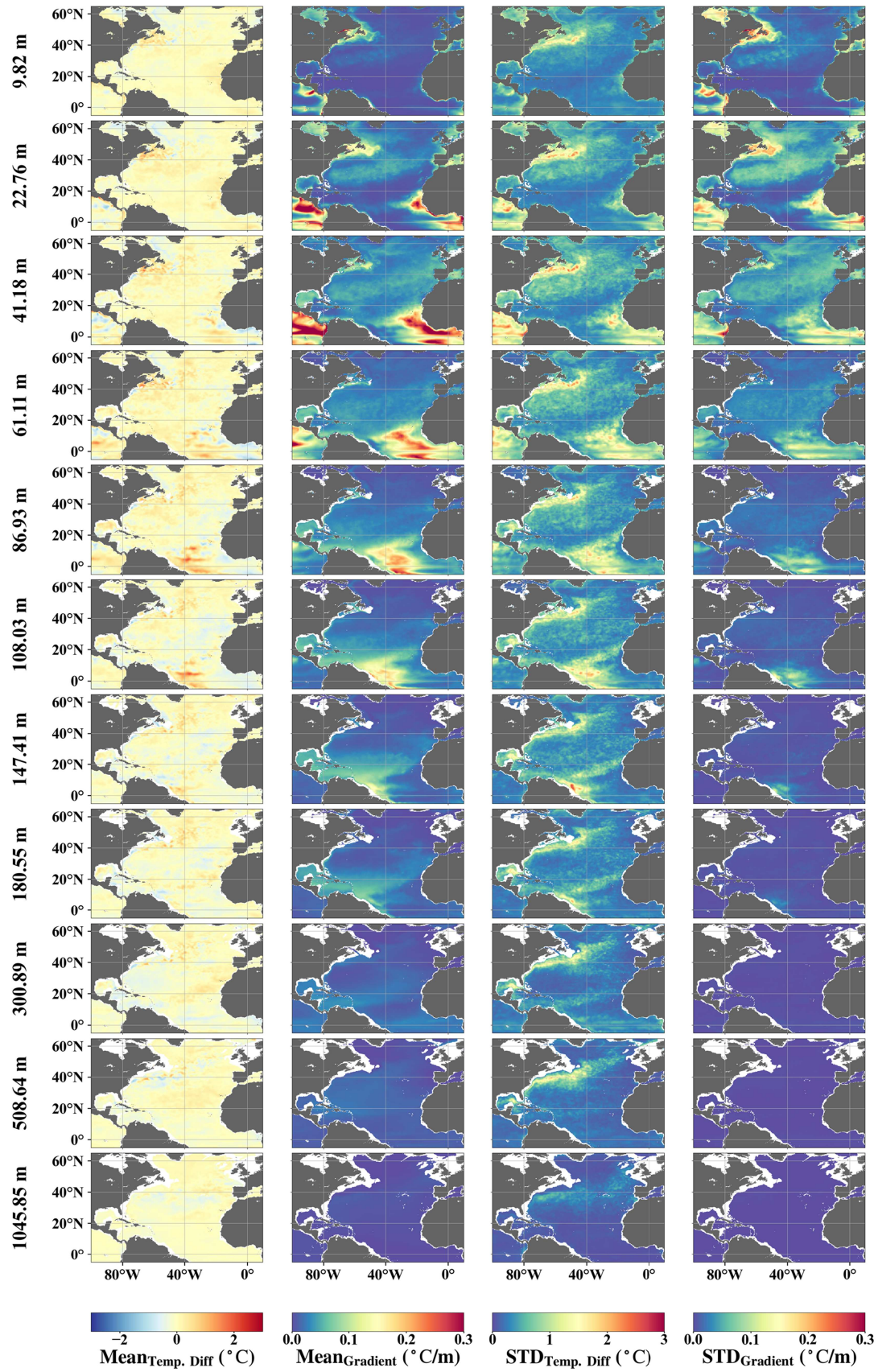
**Figure 4.** Time–depth series plots of the bias, STD, and RMSE of the ocean temperature reconstruction errors for multiple feature combination models. The solid black line represents the time series variation of the MLD.

gradient and rapid water temperature changes. Regarding bias, the mixed layer exhibited a relatively stable positive bias throughout the year, indicating that the model tended to overestimate the upper ocean temperature. The bias in the upper thermocline exhibited significant seasonality, including a significant negative bias from August to November, indicating that the model underestimated the temperatures during this period, whereas a positive bias from March to July indicated that the model overestimated the temperatures. This seasonal variation in bias is likely closely related to the seasonality of external climate forcing, such as solar radiation and wind speed, as well as intensity variations within the thermocline. All feature combinations showed similar seasonal characteristics in RMSE and STD but differed slightly in terms of bias. Combinations, including SLA, showed a persistent positive bias in the mixed layer, whereas other feature combinations occasionally turned negative, particularly from August to October 2020.

To evaluate the spatial performance of the model, Figure 5 shows the spatial distribution of the RMSE of the best-performing model using SST, SSH, SSW, and GRAD as inputs. In the upper ocean, particularly in the 0–100-m depth range, the most significant errors were concentrated in the Gulf Stream region. This is physically expected because it is one of the most dynamically complex ocean regions globally, characterised by high-velocity currents, strong heat advection, and significant instabilities such as meanders and eddy shedding (Watts and Johns 1982; Zhong and Bracco 2013). These factors make the relationship between the surface and subsurface data extremely nonlinear and variable. Significant errors also appeared in the tropical subsurface region at depths of 40–200 m, which shifted westwards with increasing depth. This pattern may be related to heat transport and pathways of the North Equatorial Current (Kanzow, Send, and McCartney 2008). Below 200 m, the errors decreased significantly across the entire region. Most of the remaining error is concentrated in mid-latitude temperate regions, whereas the error in the deeper sea is



**Figure 5.** Spatial distribution maps of the RMSE of the reconstructed ocean temperature compared with the reference values at different depths.



**Figure 6.** Spatial distribution of the bias and STD of reconstruction ocean temperature error, as well as the bias and STD of vertical temperature gradient at different depth levels.

minimal. Overall, the spatial error characteristics show consistent patterns in key dynamic regions, such as the Gulf Stream and equatorial system.

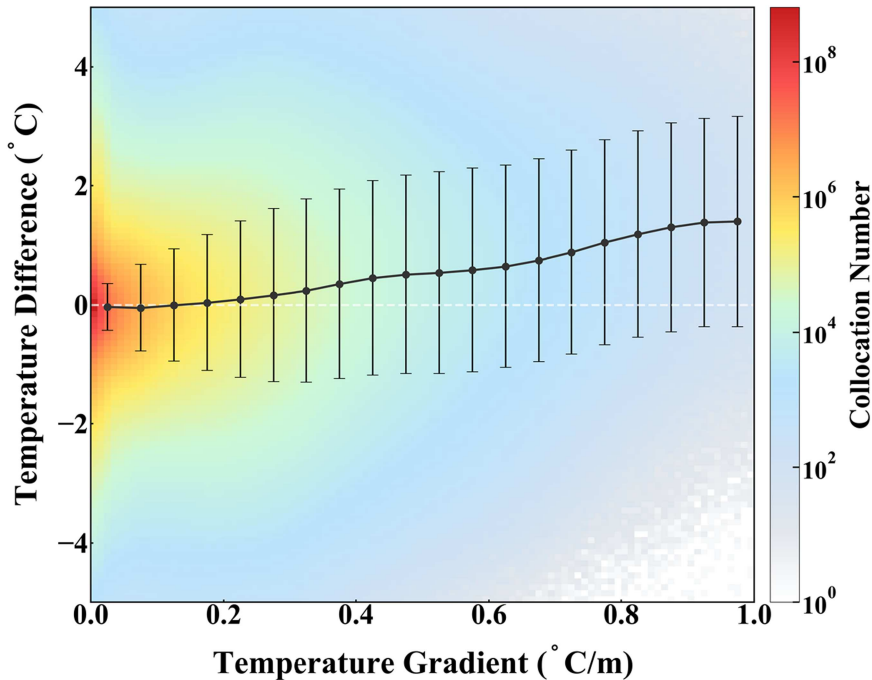
## 5. Discussion

### 5.1. Spatial errors with vertical temperature gradient

The vertical temperature gradient is a crucial physical indicator for measuring the stratification phenomenon in the ocean and intensity of the thermocline. To verify whether the model could effectively capture the vertical correlation of sea temperature, we analysed the correlation between the reconstruction error and vertical temperature gradient.

Figure 6 shows the spatial consistency between the model error and vertical gradient strength within different depth ranges. In regions with strong stratification characteristics, such as the Gulf Stream and tropical thermocline areas, the model exhibited a higher reconstruction bias and STD. This spatial distribution consistency indicates that the model actively learns the complex nonlinear vertical structure. However, its performance is physically constrained by the inherent difficulty in resolving sharp vertical gradients.

Figure 7 further quantifies this connection through the point density relationship between the error and vertical temperature gradient. As the vertical temperature gradient increases, the bias shows a significant upward trend and deviates from zero. Meanwhile, the STD of the error also significantly expands. This trend confirms that the model behaves very consistently physically, maintaining high accuracy in the vertically uniform regions but showing errors in regions with a larger gradient. This indicates that the model error is not randomly distributed but concentrated in high-gradient regions where complex thermocline changes are difficult to precisely describe. The model has learned the implicit spatial and vertical correlations, but this correlation relies on the statistical laws derived from the training data and lacks clear physical constraints to strictly ensure continuity. In regions with intense temperature changes, this statistical constraint often leads to smoothing processing of the reconstruction results, losing certain details.

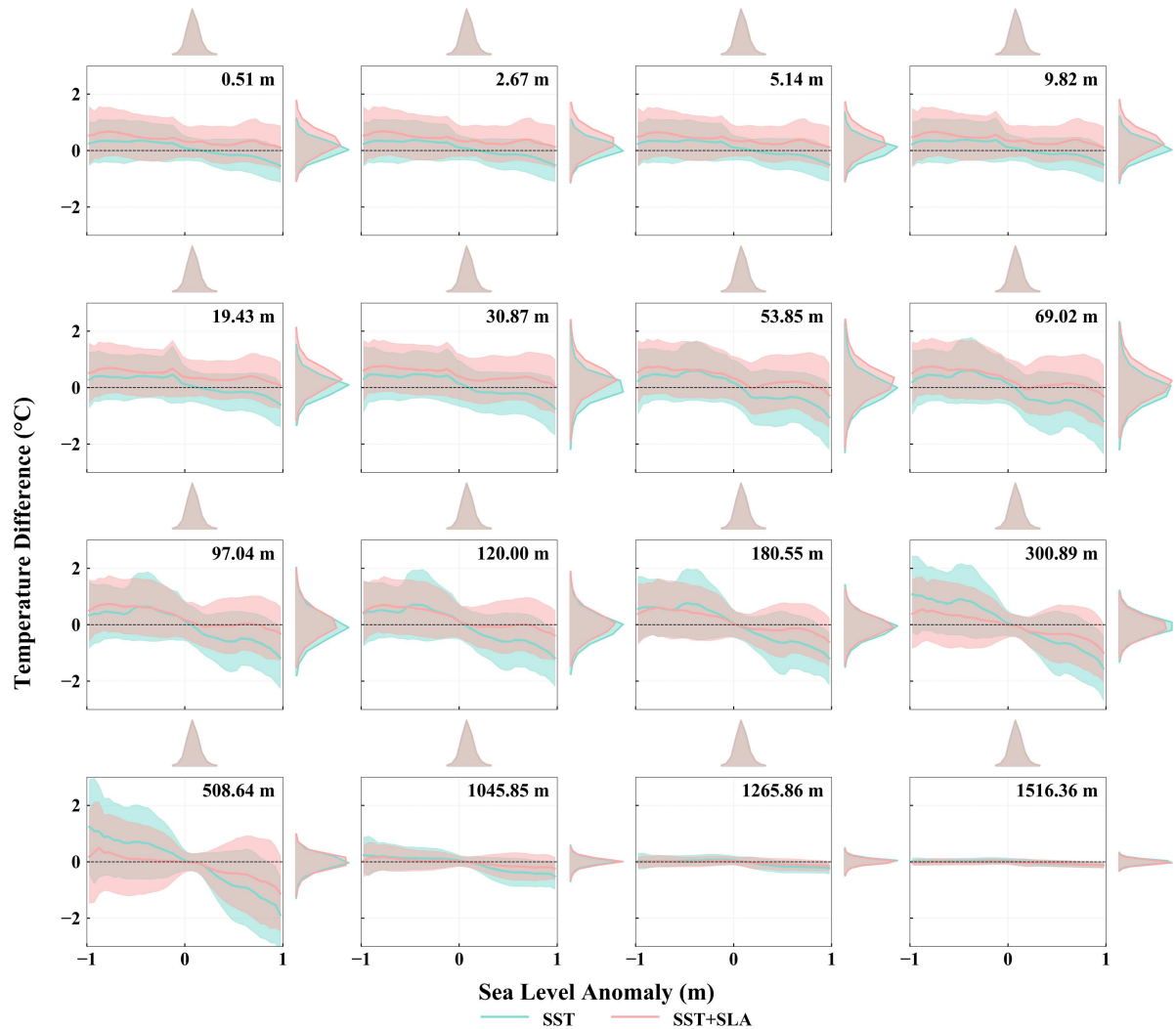


**Figure 7.** Relationship between reconstructed ocean temperature error and vertical temperature gradient. The colour represents the density of the points, the black curve represents the bias, black vertical line represents the STD, and white dashed line represents error = 0.

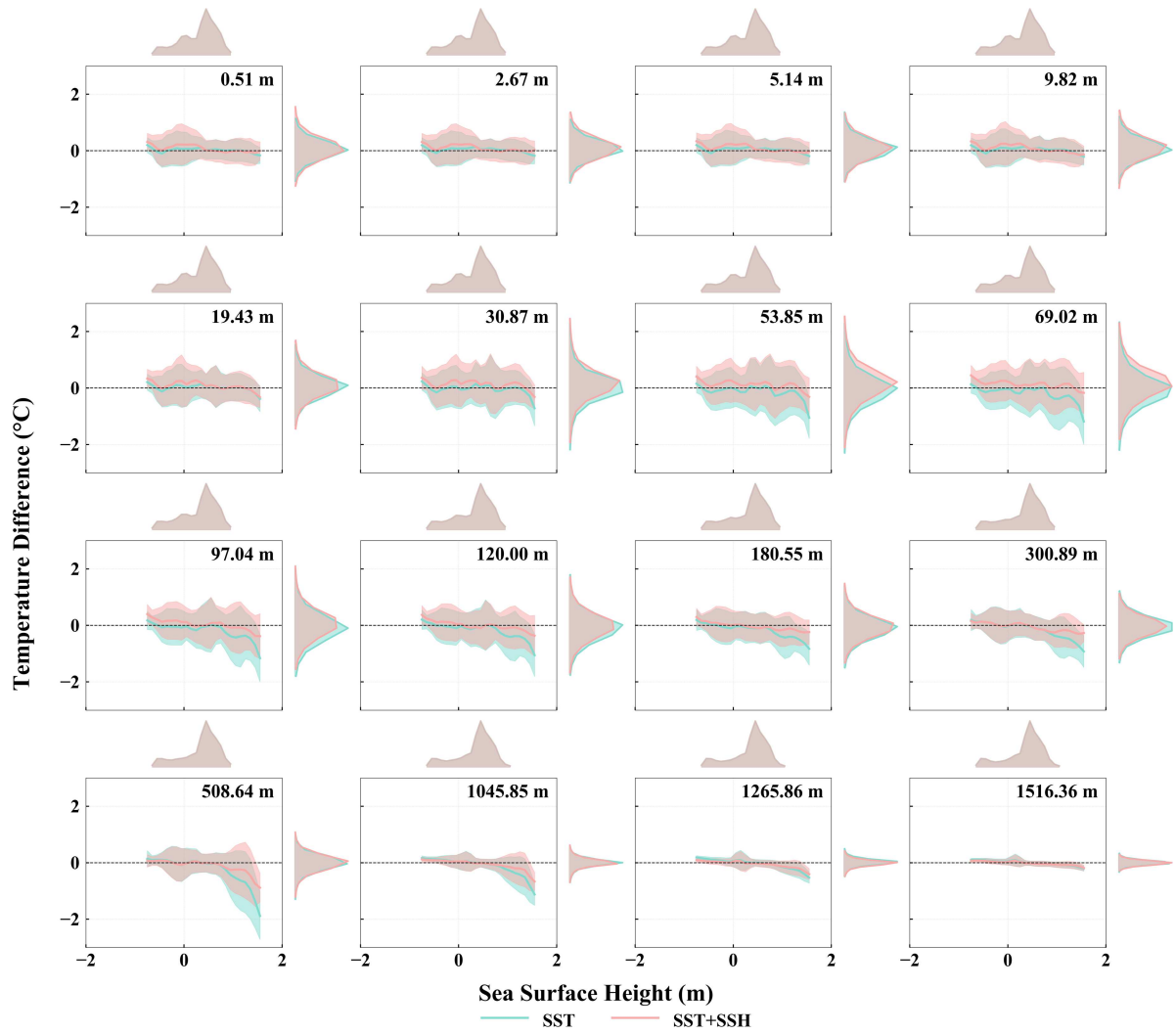
## 5.2. Correlation between error and sea surface features

To further analyse the relationship between the error and specific sea surface features, we plotted the functional relationships between the error and SLA, SSH, GRAD, and SSW at different depths, as shown in Figures 8–11. In these figures, the x-axis represents the sea surface feature value and the y-axis represents the error. The green curve represents the baseline model using only SST, and the red curve represents the model incorporating the corresponding sea surface features. The main curve represents the bias and the shaded area represents the STD.

Figure 8 shows that the SLA exhibits the strongest correlation with the model error. The baseline model shows a strong inverse bias: when the SLA is negative, corresponding to cold eddies or dynamic lows, the model tends to overestimate the temperature, resulting in a positive bias; when the SLA is positive, corresponding to warm eddies or dynamic highs, the model tends to underestimate the temperature, resulting in a negative bias. When the SLA is added to the model, as shown by the red curve, this bias trend flattens significantly. Particularly, at depths of 0–50 m, adding SLA increases the overall bias, but below 50 m, it effectively reduces the bias, bringing it closer to 0 °C. - Figure 9 shows that the effect of SSH was less pronounced. The baseline model exhibited a significant negative bias in high-SSH regions, starting below 30 m and continuing up to approximately 1200 m. Incorporating the SSH feature effectively mitigated this



**Figure 8.** Relationship between reconstructed ocean temperature error and SLA before and after including SLA in the feature model. Green represents the model values without SLA, and red represents the model values with SLA. The curves represent the bias, and the shaded areas represent the STD. The curves at the top and right represent the histogram distributions of SLA and errors, respectively.



**Figure 9.** Relationship between reconstructed ocean temperature error and SSH before and after including SSH in the feature model. Green represents the model values without SSH, and red represents the model values with SSH. The curves represent the bias, and the shaded areas represent the STD. The curves at the top and right represent the histogram distributions of SSH and errors, respectively.

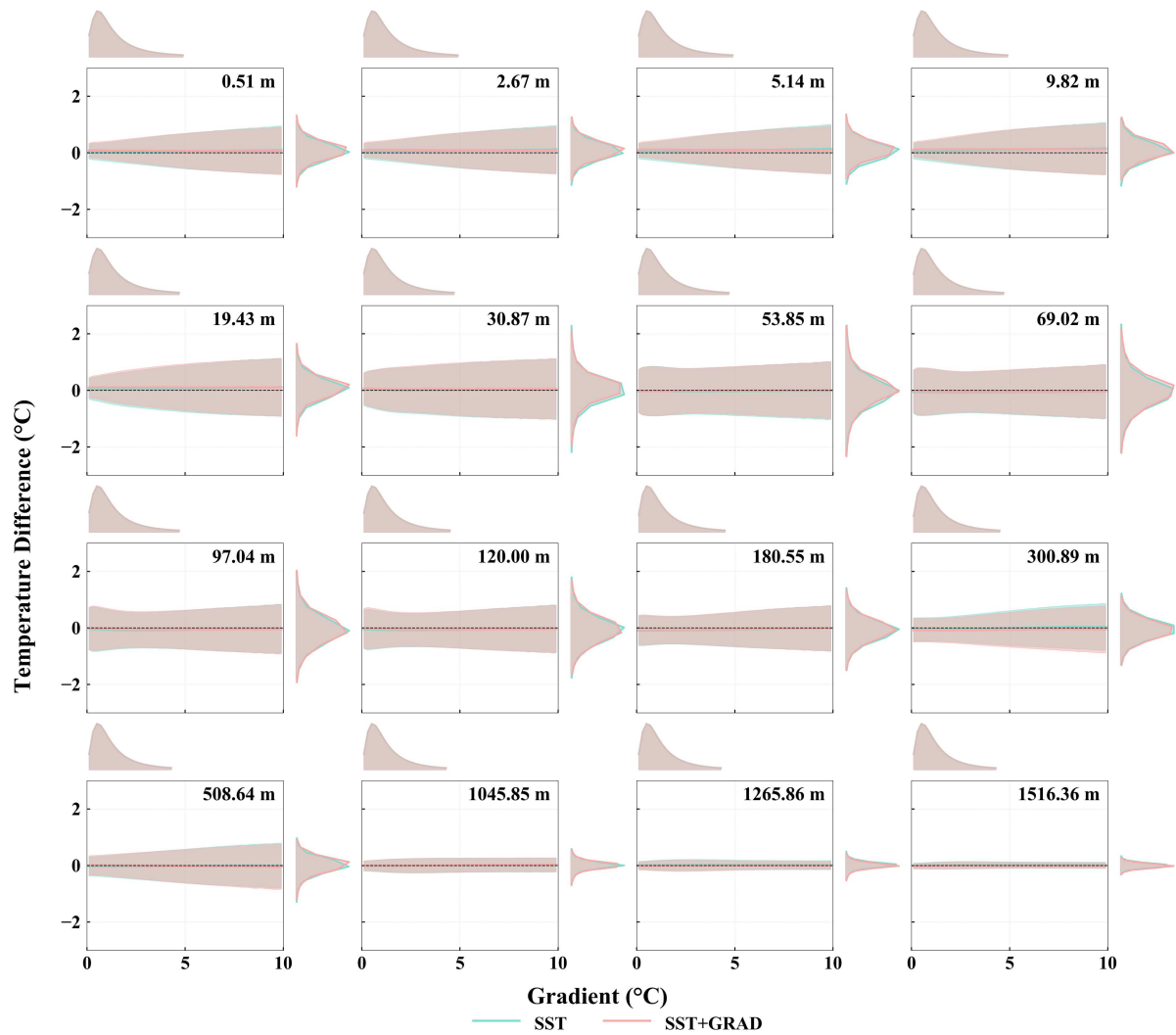
negative bias in high-SSH regions. In contrast, GRAD (Figure 10) and SSW (Figure 11) show weaker correlations with error bias. Their main contribution appears to be a slight reduction in the STD of the error, with this effect primarily concentrated in shallower layers above 50 m.

Notably, in all figures, as the sea surface feature values on the x-axis approach the extremes of their distribution, the data sample size, as shown in the top histogram, decreases sharply, reducing the representativeness of the error estimates in these extreme regions. Similarly, as the ocean temperature stabilises with increasing depth, the sample size decreases and the impact of sea surface features on the error weakens. Overall, the introduction of sea surface features played a role in reducing bias. Furthermore, the histograms show that the accuracy of the model was relatively good within the data distribution range.

### 5.3. SHAP analysis

#### 5.3.1. Vertical structure of feature contributions

We used SHAP analysis to quantify the relative importance of the five sea surface features in the reconstruction. Figure 12 shows the variation in the absolute SHAP value with depth, and Table 5 presents

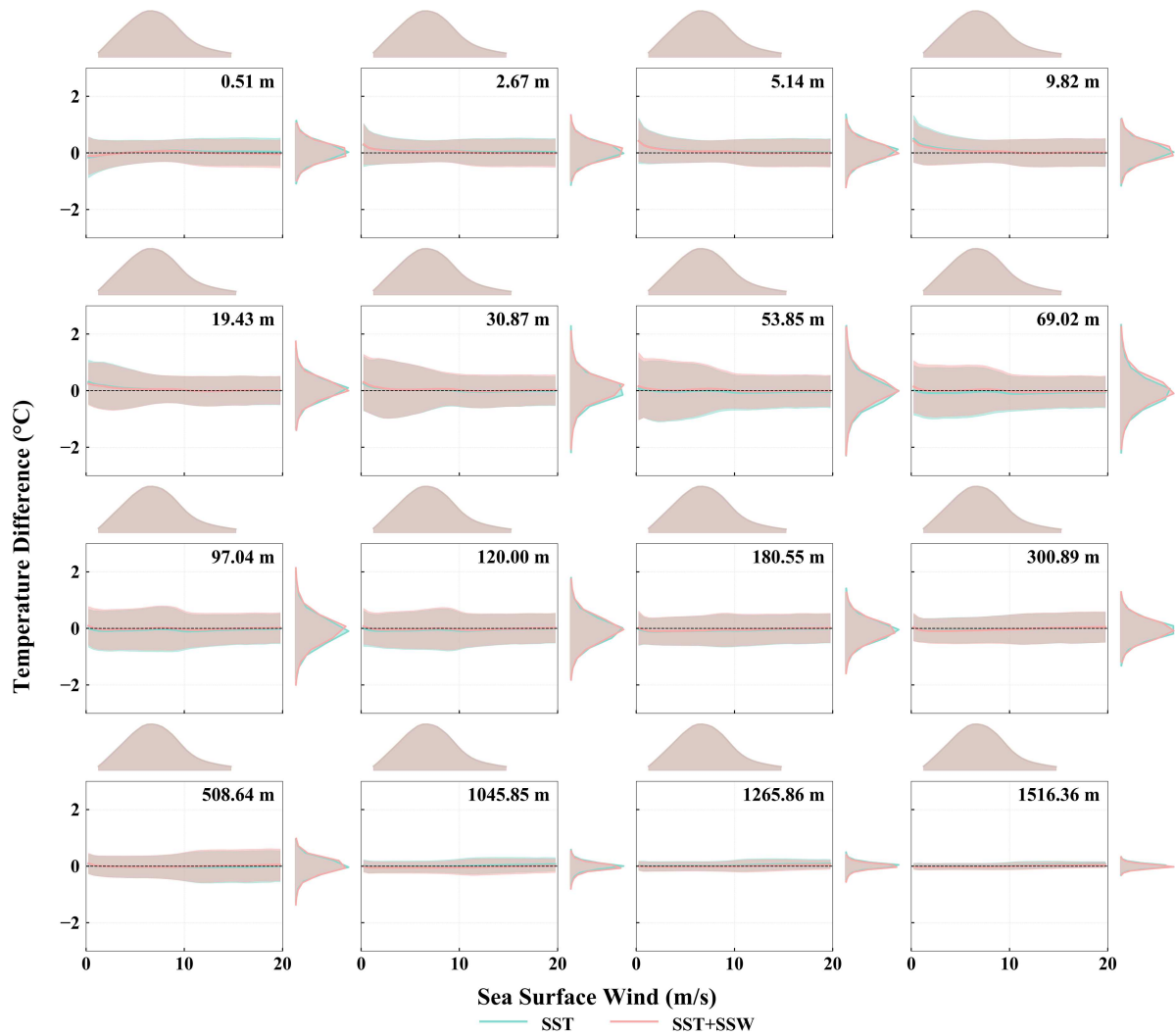


**Figure 10.** Relationship between reconstructed ocean temperature error and GRAD before and after including GRAD in the feature model. Green represents the model values without GRAD, and red represents the model values with GRAD. The curves represent the bias, and the shaded areas represent the STD. The curves at the top and right represent the histogram distributions of GRAD and errors, respectively.

a summary of their relative contributions in the mixed layer, thermocline, and deep layer. At the sea surface (0.5 m), the reconstruction was mainly controlled by SST, SSW, and GRAD. SST had the greatest influence, with a SHAP value close to 0.7 °C; SSW and GRAD were next, at approximately 0.4 °C; SLA and SSH had the smallest influences, at approximately 0.3 °C and 0.25 °C, respectively. In the mixed layer, the influence of internal ocean dynamics and vertical heat transport increased with depth. The contributions of SST, SSW, and GRAD decreased, whereas those of the dynamic features, SSH and SLA, increased. At the bottom of the mixed layer, the SHAP value of SST decreased to approximately 0.4 °C, whereas SLA increased to approximately 0.35 °C. Overall, SST dominated, with a relative contribution of 34.01% in the mixed layer, followed by GRAD at 18.66% and SSW at 17.96%.

In the mixed layer, the influence of internal ocean dynamics and vertical heat transport increased with depth. The contributions of SST, SSW, and GRAD decreased, whereas those of the dynamic features, SSH and SLA, increased. At the bottom of the mixed layer, the SHAP value of SST decreased to approximately 0.4 °C, whereas SLA increased to approximately 0.35 °C. Overall, SST dominated, with a relative contribution of 34.01% in the mixed layer, followed by GRAD at 18.66% and SSW at 17.96%.

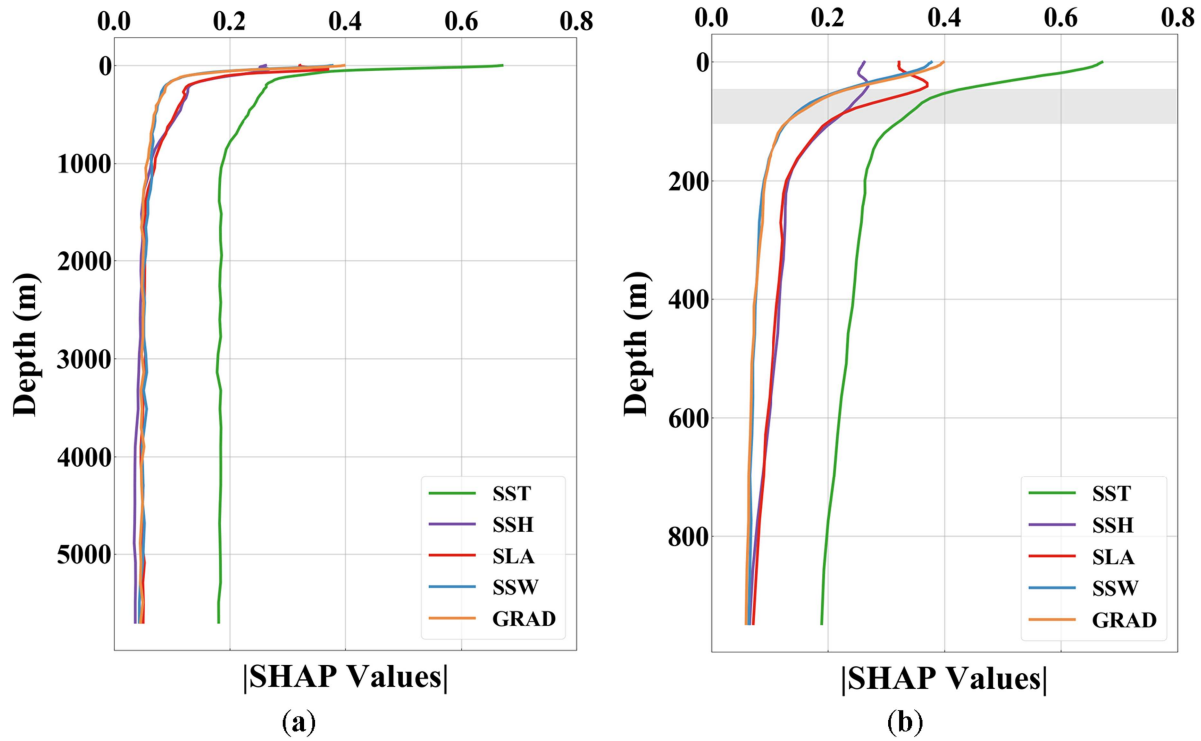
Importance of dynamic feature peaks in the thermocline. SST remained the leading contributor at 25.34%, but the relative importance of SLA (23.09%) and SSH (18.96%) significantly increased, surpassing



**Figure 11.** Relationship between reconstructed ocean temperature error and SSW before and after including SSW in the feature model. Green represents the model values without SSW, and red represents the model values with SSW. The curves represent the bias, and the shaded areas represent the STD. The curves at the top and right represent the histogram distributions of SSW and errors, respectively.

those of SSW (15.55%) and GRAD (17.06%). This shift indicated a distinct transition in the dominant driving mechanisms across the depth layers. While SST governs the thermodynamics of the upper mixed layer, the surge in the importance of SSH and SLA within the thermocline demonstrates that the framework effectively captured the strong physical correlation between surface height and subsurface thermal structure. Physically, anomalies in SSH usually reflect the expansion or contraction of the water column below, corresponding to the deepening or shoaling of the isotherms. By leveraging these dynamic proxies, the model can infer the vertical thermal displacements caused by mesoscale dynamics, which surface thermal signals alone cannot resolve. From this layer onward, the contributions of all the features steadily decreased with depth. Nevertheless, SST was dominant, remaining above  $0.3^{\circ}\text{C}$  at the bottom of the thermocline and decreasing to  $0.2^{\circ}\text{C}$  at 800 m. SLA decreased to a level comparable to that of SSH at sea level at the bottom of the thermocline, approximately  $0.2^{\circ}\text{C}$ . SSW and GRAD decreased even more rapidly, reaching approximately  $0.15^{\circ}\text{C}$  at the bottom of the thermocline.

Below 1000 m, SSH, SLA, SSW, and GRAD were close to each other, all below  $0.1^{\circ}\text{C}$ . SST remained close to  $0.2^{\circ}\text{C}$ . Overall, an SST of 34.30% was the undisputed dominant feature at all depths. SLA ranked second with 18.10%, whereas SSH, SSW, and GRAD had similar overall contributions.



**Figure 12.** Absolute SHAP values of SST, SSH, SLA, SSW, and GRAD with respect to depth. (a) All depths. (b) Depth range of 0–1000 m.

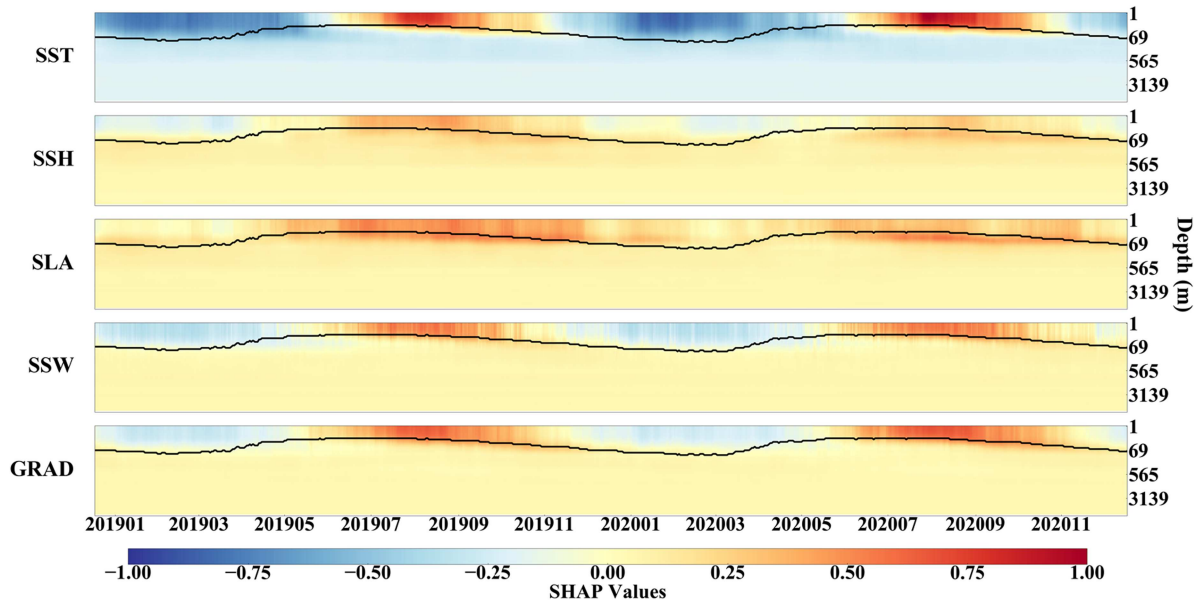
**Table 5.** Relative contribution of the absolute SHAP values of SST, SSH, SLA, SSW, and GRAD in mixed layer, thermocline, and deep layer.

| Sea surface features | SHAP <sub>Mix</sub> | SHAP <sub>Thermocline</sub> | SHAP <sub>Deep</sub> | SHAP <sub>All depth</sub> |
|----------------------|---------------------|-----------------------------|----------------------|---------------------------|
| SST                  | 34.01%              | 25.34%                      | 41.37%               | 34.30%                    |
| SSH                  | 12.97%              | 18.96%                      | 16.08%               | 15.40%                    |
| SLA                  | 16.40%              | 23.09%                      | 16.92%               | 18.10%                    |
| SSW                  | 17.96%              | 15.55%                      | 13.19%               | 15.90%                    |
| GRAD                 | 18.66%              | 17.06%                      | 12.44%               | 16.30%                    |

### 5.3.2. Seasonal and spatial variation of contributions

The feature contributions also exhibit significant seasonality, as shown in Figure 13. SST, SSW, and GRAD showed the most pronounced seasonal characteristics, primarily contributing positively in summer and autumn, leading to increased predicted temperatures, and contributing negatively in winter and spring. In contrast, SLA provided a positive contribution almost year-round, but its contribution was stronger in summer and autumn. SSH had the weakest seasonal signal.

The spatial distribution of the feature contributions shown in Figure 14 also aligns with regional physical dynamics. The contribution of SST was widely distributed in the upper ocean. The contribution of the SSH was mainly concentrated along the path of the Gulf Stream. This reflects the capability of the model to use the mean dynamic topographic pattern to locate the permanent boundaries of strong boundary currents. In contrast, the contribution of the SLA exhibited a widespread high-intensity pattern across the entire mid-latitude band. This region corresponds to an area with intense mesoscale eddy activity in the North Atlantic. The consistently high SHAP values suggest that the SLA serves as a critical indicator for identifying mesoscale eddies and transient variability. Because eddies induce significant vertical displacements of isotherms, the model identified the SLA as a significant predictor for reconstructing these complex subsurface anomalies. Despite the reconstruction challenges posed by the high nonlinearity of eddy dynamics, the pronounced contribution of the SLA confirmed that the decision-making process of the model aligns with the physical principle that surface dynamic signals are essential for inferring an ocean's internal thermal structure. The influences of SSH and SLA persist to depths of



**Figure 13.** Time–depth series of SHAP values for SST, SSH, SLA, SSW, and GRAD. The black line represents the time series variation of the MLD.

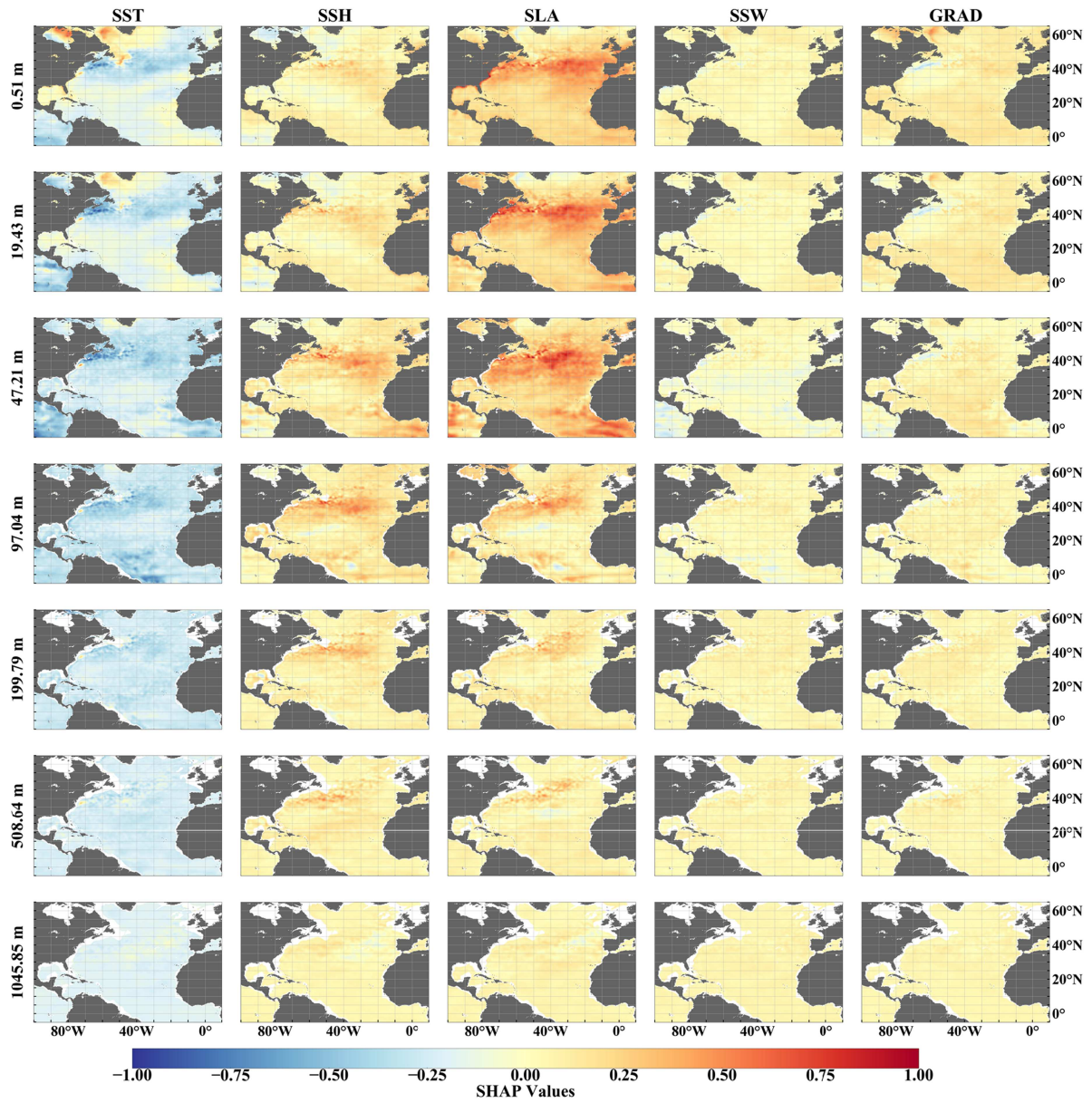
approximately 1000 m and are concentrated in the Gulf Stream region. In contrast, the contributions of SSW and GRAD are relatively small, and their influences are largely limited to shallow layers.

## 6. Conclusions

This study uses multi-source satellite data to reconstruct the 3D ocean temperature field in the North Atlantic based on a deep learning model, achieving an overall RMSE accuracy of 0.49 °C. The study revealed that the model error is not randomly distributed but is closely related to the physical structure of the ocean. Spatially, the error is concentrated in high-gradient areas such as the Gulf Stream; vertically, the error reaches its peak at the thermocline top. Temporally, the error showed strong seasonal fluctuations in the upper ocean from 0 to 200 m. Spatial, temporal, and depth-based SHAP analyses provide the physical attributes of these error patterns. SST dominates the thermodynamic process of the mixed layer, whereas SLA and SSH are the key indicators for inferring the thermocline structure. In addition to its high-precision reconstruction capability, the proposed method has significant advantages over traditional numerical models in terms of the operational efficiency. By learning end-to-end direct mapping, the model bypasses the computational bottleneck of solving complex partial differential equations (PDEs) in numerical simulations. Once trained, it can generate high-resolution 3D fields nearly in real time. This capability provides strong support for time-sensitive tasks, such as rapid ocean monitoring.

Despite these findings, our analysis reveals several limitations of the current purely data-driven methods. First, because the ocean is a non-stationary system evolving over time, models trained on historical data struggle to handle future interannual trends or long-term changes. Second, the models capture implicit spatial and vertical correlations but lacks explicit physical constraints. In high-gradient regions with intense temperature changes, these statistical regularities often fail to ensure strict continuity, resulting in smoothed reconstruction results and the loss of fine structures. Finally, although reanalysis data provides the spatially and temporally continuous gridded labels, it does have inherent biases compared to in situ observations and often exhibits a certain smoothing effect as an assimilation product.

Future research should focus on addressing these challenges and expanding the capabilities of the model. First, to address the limitations of the models, future improvements will focus on directly incorporating time–evolution mechanisms and explicit physical equations into deep learning architectures. We will explore methods such as Physical Information Neural Networks (PINNs), embedding control equations in the loss function, and integrating time-series modules such as LSTM or Transformers.



**Figure 14.** Spatial distribution maps of SHAP values for SST, SSH, SLA, SSW, and GRAD at different depths.

Simultaneously, we will consider introducing sparse but high-precision in situ observations into the training process, using a mixed training strategy to correct the model, thereby improving the accuracy of fine structure reconstruction while maintaining the advantage of full coverage. Second, based on the findings of the SHAP analysis, future reconstruction work will benefit from the development of depth-dependent feature-weighting schemes to better capture the unique physical processes followed by different depth layers. Moreover, although this study established a benchmark for the dynamic and complex North Atlantic, extending the model to a global scale remains the primary goal. Future studies should adopt spatial partitioning training strategies to address the computational challenges of global high-resolution reconstruction.

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## Author contributions

CRediT: **Fanli Liu**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft; **Li Wei**: Funding acquisition, Writing – review & editing; **Lei Guan**: Conceptualization, Funding acquisition, Investigation, Project administration, Resources, Supervision, Writing – review & editing.

## Disclosure statement

The authors report there are no competing interests to declare.

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## Data availability statement

The raw data used to construct the training and validation datasets are publicly available. The Global Ocean OSTIA Sea Surface Temperature, Global Ocean Gridded L4 Sea Surface Heights, and Global Ocean Ensemble Physics Reanalysis data were obtained from the Copernicus Marine Data Store (DOI: [10.48670/MOI-00168](https://doi.org/10.48670/MOI-00168); [10.48670/MOI-00024](https://doi.org/10.48670/MOI-00024); and [10.48670/MOI-00148](https://doi.org/10.48670/MOI-00148)). The CCMP Wind Vector Analysis Product was obtained from the Remote Sensing Systems (DOI: [10.56236/RSS-uv6h30](https://doi.org/10.56236/RSS-uv6h30)).

The processed datasets derived from the raw sources and the reconstructed 3D ocean temperature fields generated using the deep learning model are available from the corresponding author upon reasonable request.

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